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Atorvastatin might resist tobacco smoking-induced endothelial inflammation through the inhibition of NF-κB signal pathway

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ABSTRACT

Endothelial inflammation caused by tobacco smoking is widely considered as a pathogenic factor in many vascular diseases. Drugs such as atorvastatin were found to be an effective treatment in smoking-dependent vascular diseases, but the underlying mechanism is still unclear. Here, we investigated the mechanism of atorvastatin resisting endothelial inflammation caused by tobacco smoking. Firstly, isolated human umbilical vein endothelial cells (HUVECs) were divided into normal control group, cigarette smoking extract (CSE) group, and atorvastatin (AS)+CSE group. Then the expressions of inflammatory factors (vascular cell adhesion molecule-1 (VCAM-1) and E-selectin) and nuclear transcription factor kappa (NF-κB) in HUVECs were detected by western blot after separate treatments. The results showed that the expressions of VCAM-1, E-selectin, and NF-κB in CSE group were significantly higher than the other two groups ($P< 0.05$). We also found that the expressions of VCAM-1, E-selectin, and NF-κB in CSE + atorvastatin group were a little higher than the normal control group ($P< 0.05$). Our results showed that atorvastatin might partly resist tobacco smoking-induced endothelial inflammation through the inhibition of NF-κB signal pathway.

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vascular endothelial cells; tobacco smoking; nuclear transcription factor kappa B (NF-κB) signal pathway; vascular cell adhesion molecule-1(VCAM-1); E-selectin

Background

Accumulating evidences suggest that tobacco smoking is associated with many vascular diseases (1–3). Endothelial inflammation induced by tobacco smoking plays a particularly important role in the pathogenesis of vascular diseases (4–6). Although the exact mechanisms about how tobacco smoking induces endothelial inflammation are still unknown, the activation of nuclear transcription factor kappa B (NF-κB) signal pathway was supposed to be involved in this inflammatory process by upregulating the expressions of many inflammatory factors (7). Vascular cell adhesion molecule-1 (VCAM-1) and E-selectin, two important downstream target genes of NF-κB signal pathway (6,8), were proved as two important inflammatory factors for endothelial inflammation (9,10). So drugs that could decrease the endothelial inflammation might be a remarkable treatment for vascular diseases.

Atorvastatin, a widely used cholesterol-lowering drug, was found to attenuate the activation of NF-κB signaling (11,12). So we explored whether atorvastatin could inhibit endothelial inflammation by downregulating the expressions of inflammatory factors through the inhibition of NF-κB signal pathway. In this study, we detected the expression of two important endothelial inflammatory factors (VCAM-1 and E-selectin) and NF-κB in human umbilical vein endothelial cells (HUVECs) in three different groups (normal control (NC) group, cigarette smoking extract (CSE) group, and atorvastatin (AS)+CSE group) and compared the expression of VCAM-1, E-selectin and NF-κB in different groups as to further eliminate the effect of atorvastatin on endothelial inflammation and the potential mechanism.

Materials and methods

Cell and drug

HUVECs were provided by the Cell Bank of Chinese Academy of Sciences, Shanghai, China. Cells were cultured in DMEM media supplemented with 10% fetal bovine serum (FBS). Atorvastatin was provided by Pfizer Pharmaceuticals Ltd, New York, USA.

Preparation of CSE

CSE was smoked from available cigarettes from markets according to the US Federal Trade Commission (FTC) guidelines (13). CSE was smoked 35 mL per 2 seconds, once of 58-second interval until the end of cigarettes. The concentration of CSE was one cigarette per mL. RPM1640 was used as the absorption liquid.

Cell grouping and treatment

Three groups were devised: (1) normal control group (NC group), (2) CSE treatment group (CSE group), and (3) atorvastatin (AS) and CSE treatment group (AS+CSE group). NC group was normal HUVECs without any treatment, CSE

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group was treated with CSE at a concentration of 10%, AS +CSE group was first treated with atorvastatin at a concentration of 10 μmol/L for 4 h and then treated with CSE at a concentration of 10%. All the cells were cultured for 12 h and then were extracted for total protein.

Western blot

Cells were lysed in Radio Immunoprecipitation Assay (RIPA) buffer containing 50 mM Tris–HCl, 150 mM NaCl, 1% NP-40, 0.1% SDS, 0.5% sodium deoxycholate, 2 mM sodium fluoride, 2 mM Na_3VO4_2, 1 mM EDTA, and 1 mM EGTA and were analyzed by western blotting (14). Samples (20 μg of total protein or from 1 mg of total cell proteins) were loaded onto SDS-PAGE gels (Invitrogen) and separated by size using electrophoresis. Proteins were then transferred to PVDF membranes for 1 h. The membranes were then blotted for 1 h with 5% milk. Membranes were incubated with primary antibodies (1:500 dilution) against VCAM-1 (Santa Cruz Biotechnology, Inc., USA), E-selectin (Santa Cruz Biotechnology, Inc., USA), and NF-κB (Santa Cruz Biotechnology, Inc., USA) at 4°C overnight. After incubation with horseradish peroxidase (HRP)-conjugated secondary antibody (1:1000 dilution) for 3 h at 37°C, proteins were detected by ECL chemiluminescence HRP substratefor 5 minutes. The films were analyzed by densitometry with image software.

Statistical analysis

Statistical Package for the Social Science software (version17.0; SPSS Concepts) was used for data analysis. Data were expressed as mean ± SD and statistically evaluated by one-way ANOVA. $P < 0.05$ was considered as a significant difference.

Results

Cellular morphology of HUVECs in different groups

Drastic morphological change in HUVECs was observed in the CSE group. As shown in Figure 1, HUVECs changed from normal spindle shape to spherical shape with vacuolization, and many necrotic cells were observed. Morphological change of HUVECs in the AS+CSE group was not as much obvious as in the CSE group. Only partial HUVECs changed from spindle shape to spherical shape and no vacuolization and necrotic cells were observed.

Expression of VCAM-1 and E-selectin in HUVECs in different groups

As illustrated in Figure 2, the protein expression of VCAM-1 and E-selectin in the CSE group was significantly higher than that in the AS+CSE group and NC group ($P < 0.05$). Also, the protein expression of VCAM-1 and E-selectin in the AS+CSE

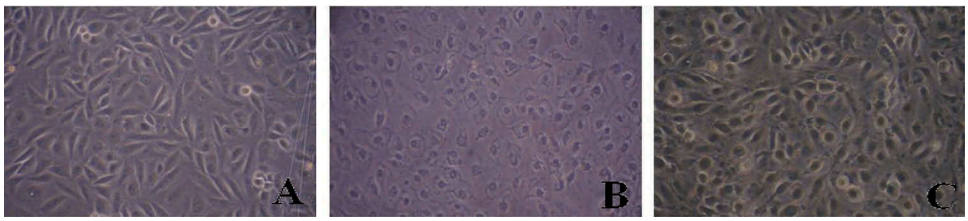

Figure 1. Microscopic examination (×200) of cellular morphology of HUVECs in different groups. Drastic morphological changes of HUVECs were observed in the CST group (B) and slight morphological changes of HUVECs were observed in the AS+CSE group (C). (A: NC group, B: CSE group, C: AS+CSE group).

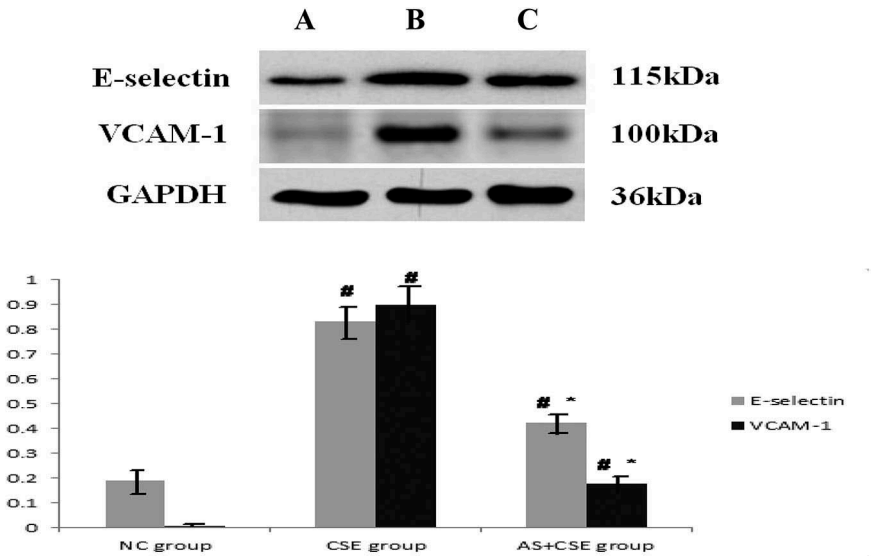

Figure 2. Western bolt shows the expression of VCAM-1 and E-selectin in HUVECs in different groups.

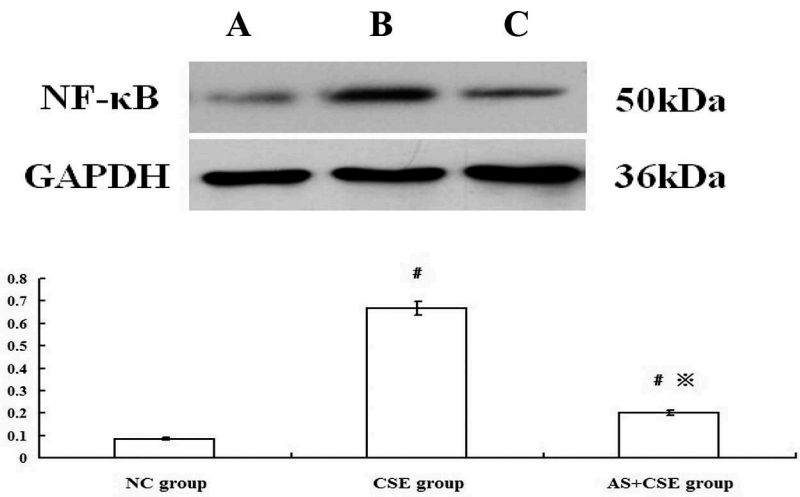

Figure 3. estern blot shows the expression of NF-κB in different groups.

group was just slightly higher than that in the NC group, but was significantly lower than that in the CSE group ($P < 0.05$).

Expression of NF-κB in HUVECs in different groups

As illustrated in Figure 3, the protein expression of NF-κB was much higher in the CSE group than that in the AS+CSE group and NC group ($P < 0.05$), The protein expression of NF-κB in the AS+CSE group was a little higher than that in NC group ($P > 0.05$), but significantly lower compared with the CSE group ($P < 0.05$).

Discussion

The risk of cardiovascular death from tobacco smoking continues increasing for the people all over the world (15). Vasomotor dysfunction, alterations in thrombosis/fibrinolysis, and modification of lipids may mediate smoking-related vascular diseases, involving a central role of vascular inflammation and oxidative stress (16). Although the pathogenesis of smoking-related vascular diseases remains incompletely understood, vascular inflammation has been confirmed as a critical pathogenic factor (4), and activation of NF-κB signal pathway is widely accepted in smoking-related vascular diseases (17,18).

NF-κB signal pathway plays an important role in inflammation conditions (19–21). NF-κB is activated by several signaling pathways, such as tumor necrosis factor α signaling, interleukin-1 receptor signaling, and Toll-like receptors signaling (22). Also, NF-κB can be activated by some other stimulus, including hypoxia (23), reactive oxygen species (24), and mechanical forces (25). Activated NF-κB can induce endothelial inflammation through activating many downstream target genes, such as VCAM-1 and E-selectin (6,8). Upregulated VCAM-1 and E-selectin, two important adhesion factors of vascular endothelial cells, can increase the adhesive capacity of vascular endothelial cells and trigger vascular inflammation (9,26). Based on the significant effect of NF-κB signal pathway in vascular inflammation, drugs which inhibit the activation of NF-κB signal pathway might be considered as an effective treatment for inflammation-related vascular diseases. Atorvastatin, a widely used cholesterol-lowering drug, was found to attenuate the activation of NF-κB signaling (11,12). So we speculate that atorvastatin might be a potential target in the treatment of smoking-related vascular diseases.

In this study, we found that the cellular morphology of HUVECs in the CSE group changed sharply and the expressions of VCAM-1, E-selectin, and NF-κB in HUVECs significantly increased in the CSE group. These data indicate that tobacco smoking might induce endothelial inflammation by upregulating the expressions of VCAM-1 and E-selectin through the activation of NF-κB signal pathway. We found that the expressions of VCAM-1, E-selectin, and NF-κB in HUVECs were significantly lower in the AS+CSE group compared to the CSE group, although only slight changes in cellular morphology of HUVECs were observed. This indicates that smoking-related endothelial inflammation was reversed by atorvastatin, which may be conducted through the inhibition of NF-κB signal pathway. At last, we found that the expressions of VCAM-1, E-selectin, and NF-κB in HUVECs in the AS+CSE group were still a little higher compared to the NC group. The data indicate that blocking of NF-κB signal pathway by atorvastatin might be incomplete. More studies are needed for further understanding the comprehensive mechanism involved.

In conclusion, our study shows that tobacco smoking involves the pathogenesis of vascular diseases by triggering endothelial inflammation. Activated NF-κB signal pathway takes part in the process of endothelial inflammation, and atorvastatin might inhibit tobacco smoking-induced endothelial inflammation through the inhibition of NF-κB signal pathway.

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